Parsek Roadmap

Philosophy

Parsek is a **git-like recording system** for KSP missions. You record flights sequentially, commit them to a timeline, and they play back automatically as ghost vessels while you fly new missions. There is one timeline, recordings are immutable commits, and the game always moves forward.

Like git, you can go back to any earlier point and start new work. Existing recordings don't change — they play out as ghosts alongside your new missions. There is no branching, no state reversal, and no paradoxes. Conflicts are prevented through resource budgeting: committed recordings have already claimed their costs, so the player can only spend what's actually available.

Time travel paradoxes are avoided by two invariants: **causality** (events are always processed in time-axis order) and **additivity** (the timeline is append-only — committed recordings and game actions can be added but never deleted). The world state at any point in time is fully determined by the ordered sequence of committed events before it. When a new event is inserted into the timeline (e.g., a recording committed at an earlier UT after a rewind), all derived resource values — available science, funds, reputation — are recalculated from scratch across the full timeline. Events themselves are immutable; the computed state that flows from them is not stored but recomputed whenever the event set changes.

Completed

v0.3 — Foundation

Initial release. The core loop shipped end-to-end: record a flight, commit it, rewind, see it play back as a ghost alongside new missions.

Recording & playback: Position recording with adaptive sampling and geographic coordinates. Kinematic ghost playback with opaque vessel replicas. Orbital recording with analytical Keplerian orbits. Auto-recording on launch and EVA. Chained recordings across vessel lifecycle events (undock, EVA, dock). Context-aware merge dialog (Merge to Timeline / Discard). Vessel persistence with deferred spawn, crew reservation, and resource deltas.

Ghost visual fidelity: Comprehensive part event types recorded and replayed on ghost vessels — decoupling, staging, parachutes, engines (with FX), solar panels, antennas, radiators, landing gear, lights, cargo bays, fairings, RCS (with FX), docking/undocking, inventory parts.

Career mode: Milestones (tech research, part purchases, facility upgrades, contracts, crew changes) captured as immutable timeline commits. Resource budget computed on-the-fly from recordings + milestones. Epoch isolation to prevent abandoned timeline branches from leaking. Resource deduction on revert so KSP's top bar reflects available resources. Action blocking to prevent re-researching committed tech or re-upgrading committed facilities. Per-recording rewind saves with quicksave-based timeline restoration. Rewind UI with confirmation dialog, resource reset, crew re-reservation, and action replay. Fast-forward to advance past committed recordings.

Recordings Manager UI with sortable columns, per-recording loop/hide, rewind/fast-forward, and status indicators.

Design: `docs/dev/done/design-restore-points.md`, `docs/dev/done/design-going-back-in-time.md`

v0.4 — Visual Fidelity & Polish

Orbital rotation fidelity, watch camera, heat effects, KSC scene playback, UI polish, and code refactoring.

Orbital rotation: Ghost vessels in orbital segments preserve their recorded attitude instead of always facing prograde. Rotation stored relative to the orbital velocity frame at the on-rails boundary and reconstructed from the Keplerian orbit. SAS-locked orientations hold correctly throughout the orbit. PersistentRotation mod detected and supported — spinning vessels reproduced during playback.

Watch camera: Camera follows a ghost vessel during playback. Automatically recenters on visible parts after separation events.

Ghost visuals: Re-entry heating FX with mesh-surface fire particles. Heat shield ablation, smoke and spark FX on decouple/destroy. Fairing ghost visuals. EVA kerbal facing fix. Explosion visual effect on impact with camera hold.

KSC playback: Ghosts visible in KSC scene with overlap support and distance-based part event culling.

UI & usability: Per-save settings panel (auto-record, sampling thresholds). Recording stats (max altitude, max speed, max G, distance, duration). Non-revert "Commit Flight" button. Two-phase parachute deploy. Auto-loop with per-recording interval controls. Recording groups with multi-membership. Time warp visual cutoffs and deferred spawn queue. Context-aware rewind button.

Code health: Major refactor reducing coupling between modules. Sandbox rewind fix.

Design: `docs/dev/done/design-orbital-rotation.md`, `docs/dev/done/design-camera-follow-ghost.md`

v0.5 — Recording System Redesign

Comprehensive redesign of the recording and playback systems. Multi-vessel sessions, ghost chain paradox prevention, spawn safety, time jump, rendering zones, and ghost visual hardening.

Design: `docs/parsek-flight-recorder-design.md` (consolidated design document)

Recording model:

- Segment boundary rule — only physical structural separation creates new segments; controller changes, part destruction without splitting, and crew transfers are SegmentEvents within a continuing segment
- Crash coalescing — rapid split events grouped into single BREAKUP events
- Environment taxonomy — classification (Atmospheric, ExoPropulsive, ExoBallistic, SurfaceMobile, SurfaceStationary) with hysteresis
- TrackSections — typed trajectory chunks tagged with environment, reference frame, and data source
- Reference frames — ABSOLUTE for physics, ORBITAL_CHECKPOINT for on-rails, RELATIVE for anchor-vessel proximity

Multi-vessel sessions:

- Recording tree (DAG) — multi-vessel missions recorded as a single unit with split/merge tracking across undock, EVA, dock, joint break, and breakup events
- Background vessel recording — all vessels in the physics bubble sampled at proximity-based rates with full part event capture
- Booster/debris recording — automatic tree promotion on staging with debris TTL
- Highest-fidelity-wins merge — overlapping Active/Background/Checkpoint data merged per vessel
- Flag planting recording and playback

Ghost chain paradox prevention:

- Committed recordings that interact with pre-existing vessels cause those vessels to become ghosts from rewind until the chain tip resolves
- Intermediate spawn suppression — multi-link chains only spawn at the tip
- Ghost conversion — real vessels despawned and replaced with ghost GameObjects during chain windows

- PID preservation — chain-tip spawns preserve the original vessel's persistent ID for cross-tree chain linking
- Ghosting trigger taxonomy — structural events, orbital changes, and part state changes trigger ghosting; cosmetic events do not

Spawn safety:

- Bounding box collision detection — spawn blocked when overlapping with loaded vessels
- Ghost extension — ghost continues on propagated orbit/surface past recording end while spawn is blocked
- Trajectory walkback — for immovable blockers, walks backward along recorded trajectory to find a valid spawn position
- Terrain correction — surface spawns adjusted for terrain height changes between recording and playback
- KSC exclusion zone — 50m safety radius around launch pad and runway
- Spawn-die-respawn prevention (3-cycle limit), spawn abandon flags, retry limits (150 frames)
- Snapshot situation correction (FLYING→LANDED for landed terminals, FLYING→ORBITING for orbital terminals)
- Non-leaf tree recording spawn suppression, unsafe snapshot situation blocking
- Dead crew protection on spawn (strip dead crew, abandon all-dead spawn)

Time jump:

- Relative-state time jump — discrete UT skip preserving rendezvous geometry across ghost chain windows
- Epoch-shifted orbits — orbital elements recomputed at the new UT from state vectors

Rendering & performance:

- Distance-based zones — Physics, Visual, Beyond — with zone-aware playback and part event gating
- Ghost soft caps — configurable thresholds with priority-based despawning, reduced fidelity (75% renderer culling), simplified orbit-line mode
- Terminated chain early-out — fully-terminated trees and chains skip per-frame evaluation

Ghost visual hardening: Variant textures and materials (TEXTURE/MATERIAL/GAMEOBJECT rules), damaged wheel filtering, fairing meshes with procedural truss and nosecone caps, SRB nozzle glow (FXModuleAnimateThrottle), engine shrouds with variant awareness, initial state seeding for all 16 tracking set types. Compound part visuals (fuel lines, struts). Plume and smoke trail fixes (KSPParticleEmitter native control, emission module disabled). Control surfaces, robotics/servo detection, cabin lights, animation-based deployables. RCS debounce (8-frame threshold). Part separation smoke/spark FX. Lingering particle systems on ghost despawn.

Ghost world presence: CommNet relay via antenna specs with correct combination formula. Ghost labels. ProtoVessel-based map presence — tracking station entries, orbit lines, navigation targeting, ghost icon click popup (Set Target / Watch / Focus). Harmony guard rails (27 checks across 9 files, 4+ patches). Orbit segment updates on SOI transitions. Tracking station NRE protection.

Real Spawn Control: Proximity-based UI for warping to when nearby ghost craft become real vessels. Per-craft warp buttons, sortable columns, countdown display, screen notifications. Countdown column in Recordings Manager.

Per-phase looping: Chain recordings split at atmosphere/altitude/SOI boundaries. Tree recordings split by optimizer at environment boundaries. Auto loop range trims boring bookends. Debris loop sync via LoopSyncParentIdx.

Watch mode hardening: Zone-hide exemption for watched ghosts (no 120km cutoff). Hold timer with retry for chain transitions. Auto-follow through tree branching with PID-matched descent. FF watch transfer. Camera bridge anchors at cycle boundaries. Distance guard (100km rendering-safe limit). Orbital recording exemption from distance cutoffs.

Rewind/spawn hardening: Future vessel strip (PRELAUNCH and all types via quicksave PID whitelist). Spawn state reconciliation after strip. EVA vessel removal with crew rescue. Localization key resolution at all comparison sites. Spawn position from snapshot (not trajectory endpoint). EVA endpoint spawning. Ladder state stripping. Orbital vessel spawn via state vector orbit construction.

UI: Fast-forward redesigned as instant UT jump. Simplified merge dialog. Compressed Status column (merged countdown). Group header aggregate stats. Recording group unique naming. In-game test runner (90 tests, Ctrl+Shift+T).

Code health: GhostPlaybackEngine extraction (zero Recording references, IPlaybackTrajectory interface). ChainSegmentManager extraction. ParsekPlaybackPolicy event-driven architecture. Comprehensive log audit (92% output reduction). 4600+ xUnit tests. PartStateSeeder unification. GhostBuildResult consolidation.

v0.6 — Game Actions System

Full redesign of milestone capture, resource budgeting, and action replay. Standalone resource ledger tracking every economic event on the timeline.

Design: `docs/parsek-game-actions-and-resources-recorder-design.md`

Architecture: Sidecar files capture raw data during flight; extraction to ledger on commit. Two-phase recalculate+patch on warp exit. Unified recalculation walk from UT=0 with two-tier module dependency ordering. KSP UI patched to display available funds (not gross balance).

Resource modules (8):

Module	Key mechanics
Science	Immutable awarded values + derived effective values, per-subject caps via full recalculation sorted by UT
Funds	Seeded balance from save, reservation system prevents overspending across rewinds, vessel build costs as recording-associated spendings
Reputation	Non-linear gain/loss curve, nominal vs effective values
Milestones	Once-ever binary cap, chronological priority, first-tier feeds funds and reputation
Kerbals	IResourceModule participant, reservation from UT=0 as continuous block, temporary vs permanent loss, replacement chains, stand-in generation, managed-kerbal filtering
Facilities	Upgrade/destruction/repair schemas, visual state management during fast-forward, KSP state patching, deferred replay for missing facilityRefs
Contracts	UT=0 reservation, accept/complete/fail/cancel lifecycle, parameter progress tracking, deadline failure detection
Strategies	UT=0 reservation, transform layer between first-tier and second-tier modules, commitment rates

Paradox prevention: No-delete invariant, spending reservation (science/funds), UT=0 reservation (kerbals/contracts/strategies). Milestone path qualification. Budget deduction clamping. Chain gap closure for game state events.

Crew lifecycle: Crew reservation via internal managed state (Harmony patches for KSP roster validation and astronaut complex count). Crew swap in both flight and KSC spawn snapshots. Orphaned crew rescue after vessel strip. Dead crew spawn protection. Kerbal rescue detection. Managed-kerbal event filtering.

Validated across all game modes: Sandbox (no resources), Science (science-only), Career (full complexity). 4621 tests passing.

v0.7 — Timeline

Unified chronological view of all committed career events, replacing the Game Actions window.

Design: `docs/parsek-timeline-design.md`

Timeline window: Flat chronological list of career events sorted by UT. Current-UT divider separates past (full color) from future (dimmed). Two significance tiers: Overview (mission structure — launches, milestones, contracts, facilities) and Detail (all resource transactions). Three source filters: Recordings, Actions (deliberate player choices), Events (gameplay consequences). Vessel-level telemetry (part events, segment events) stays in the Recordings Manager.

Entry display: EVA-aware (`EVA: Jeb from Mun Lander (MET 5s)` , `Board: Jeb (Mun Lander)`). Spawn at EndUT with VesselSituation (`Spawn: Vessel (Landed on Mun)`). Mission Elapsed Time on launches (KSP calendar). Humanized science subjects, tech nodes, milestones, strategy names. Color-coded: green = earnings, red = penalties, light blue = player actions, white = recordings.

Operations: Rewind (R) and Fast-Forward (FF) buttons on recording entries. GoTo cross-link opens Recordings Manager, unhides recording, expands parent groups, scrolls to target.

Data model: 26 entry types (2 recording lifecycle + 23 game actions + 1 legacy). `TimelineBuilder` with 3 collectors + stable UT sort. `IsPlayerAction` classification. Game-mode aware (career/science/sandbox). 4870 tests passing (53 timeline-specific).

Location context: Recordings become location-aware — body, biome, situation, and stock launch site name captured at recording start and end. Timeline shows location-enriched entries ("Launch: Vessel from Launch Pad on Kerbin", "Spawn: Vessel (Landed at Midlands on Mun)"). Sortable Site column in Recordings Manager with UT tiebreak. Prerequisite for logistics routes and async multiplayer.

v0.8 — Resource Snapshots & Recording Optimization

The logistics-prerequisite release line. Phase 11 and Phase 11.5 shipped across `v0.8.x` , adding resource/inventory/crew manifests to recordings and the observability/optimization work needed before long-lived looped transport routes.

Phase 11 — Resource snapshots: recordings now capture start/end resource manifests, inventory manifests, crew manifests, and dock-target identity at the boundaries needed for future route endpoint and delivery logic.

Phase 11.5 — recording optimization & observability: storage/perf diagnostics are now exposed in-game, Flight ghost LOD is live, and compact trajectory/snapshot sidecars plus readable mirrors cut comparable recording payload size by `85.17%` .

Phase 11: Resource Snapshots

Recordings capture physical resource manifests at recording start and end.

Resource manifests (implemented): `ExtractResourceManifest` walks vessel snapshot PART > RESOURCE nodes, sums by resource name. `StartResources / EndResources` fields on Recording, serialized as additive RESOURCE_MANIFEST ConfigNode. Captured at 8 boundary sites (recording start/stop, chain boundaries, breakup, background splits, optimizer splits/merges). EC and IntakeAir excluded (environmental noise). Dock target vessel PID (`DockTargetVesselPid`) captured at dock boundaries for route endpoint identification. Hover tooltip in Recordings Manager shows per-resource start-to-end with delta.

Inventory manifests (implemented): KSP 1.12 `ModuleInventoryPart` items (stored parts in cargo containers). `ExtractInventoryManifest` walks MODULE > STOREDPARTS > STOREDPART nodes. `InventoryItem { count, slotsTaken }` struct + vessel-level `totalInventorySlots` . Same capture sites as resources.

Crew manifests (implemented): Crew composition by trait (Pilot/Scientist/Engineer/Tourist). Route delivery uses generic kerbals (separate from crew reservation system). Same capture pattern.

Phase 11.5: Recording Optimization & Observability

Optimization pass before logistics routes add many long-lived looped recordings. A long career with dozens of missions will accumulate significant disk, memory, and playback pressure. Phase 11.5 now has three shipped threads and one explicit follow-up:

1. observability and diagnostics so storage/perf pressure is measurable during playtests
2. playback-side optimization via the current Flight ghost LOD policy
3. recording-side optimization across both trajectory and snapshot sidecars
4. remaining follow-up: synthetic stress benchmarking/tuning for the shipped policy and formats

Observability (shipped)

Phase 11.5 shipped the measurement/reporting pieces needed to make optimization work evidence-based:

- **Per-save storage report** — total disk size of Parsek data (sidecar files + `.sfs` metadata), broken down by recording in diagnostics.
- **Per-recording stats** — point count, part event count, orbit segment count, sidecar file sizes (`.prec` , `_vessel.craft` , `_ghost.craft`) visible in diagnostics / recording details.
- **Playback budget visibility** — playback timings, zone behavior, and FX counts exposed through diagnostics and one-shot/rate-limited logging.
- **Memory / LOD visibility** — live diagnostics for active ghost buckets and hidden-tier shell state, plus spawn/destroy timings for recent ghost lifecycle work.

Flight Playback / LOD (shipped)

Phase 11.5 also shipped the current Flight ghost LOD policy used to keep replay density sane during playtests:

- shared internal distance thresholds (`2.3 km` , `50 km` , `120 km` , watch cutoff)
- watched ghosts inside cutoff forced to full fidelity
- unwatched reduced tier (`2.3-50 km`)
- unwatched hidden-mesh tier (`50-120 km`)
- hidden-tier shells keep logical playback alive while unloading built mesh/resources
- live diagnostics reporting for `full / reduced / hidden / watched override`

Recording Storage (shipped)

The storage-focused half of Phase 11.5 removed the biggest measured trajectory-side waste without changing visible playback contracts:

- **Authoritative section sidecars** — `v1` `.prec` files stop duplicating flat `POINT` / `ORBIT_SEGMENT` data when `TrackSections` already contain the same trajectory.
- **Ghost snapshot alias mode** — identical ghost/vessel snapshots are stored once via `ghostSnapshotMode` metadata instead of always writing duplicate `_ghost.craft` files.
- **Compact binary trajectory sidecars** — current-format `.prec` files now use header-dispatched binary `v3` with exact scalar payloads, a file-level string table, and conservative sparse defaults for stable body/career point fields.
- **Compact lossless snapshot sidecars** — `_vessel.craft` / `_ghost.craft` now write as lossless `Deflate` envelopes at the highest built-in .NET compression level while still loading legacy text snapshot files.
- **Readable mirror sidecars** — default-on `.prec.txt` , `_vessel.craft.txt` , and `_ghost.craft.txt` mirrors keep the compact binary/lossless files authoritative while preserving a human-readable debugging

path and diagnostics byte accounting.
- **Storage regression harness** — representative fixtures, mixed-format round-trips, sidecar log assertions, and scenario-writer coverage protect the new format path.

Measured outcome so far: recording sidecars are no longer dominated by text storage. In the April 13, 2026 comparable log-bundle corpus, authoritative `.prec`, `_vessel.craft`, and `_ghost.craft` files totaled `1.34 MB` versus `9.03 MB` for readable text mirrors (`7.69 MB` saved, `85.17%` smaller), and the latest bundle alone dropped from `745,079 B` to `102,845 B` (`86.20%` smaller).

Remaining Follow-Up

With trajectory-side and snapshot-side storage optimization now shipped, the remaining Phase 11.5 follow-up is validation/tuning work against larger corpora rather than another planned format pass:

- run synthetic stress benchmarking against the shipped ghost LOD/storage stack
- tune thresholds or hot spots only if diagnostics show real pressure at scale
- keep future storage changes measurement-driven instead of pre-committing to another rewrite
- defer trajectory thinning, compression, or lazy loading until the snapshot bucket is re-measured and still justifies more work

v0.9 — Rewind to Separation

Phase 12 shipped: re-fly unfinished missions after multi-controllable splits (staging, undock, EVA with 2+ controllable outputs). Full design: `docs/parsek-rewind-to-separation-design.md`. The pre-implementation spec that drove v0.9 is archived at `docs/dev/done/parsek-rewind-separation-design.md`.

- **Rewind Points at split time** — every multi-controllable split writes a KSP quicksave to `saves/<save>/Parsek/RewindPoints/<rpId>.sfs` plus a persistent-id-to-slot map captured at save time, so the post-load strip can reliably identify which vessels to replace with ghosts.
- **Unfinished Flights UI group** — a virtual, computed group in the Recordings Manager lists every committed BG-crash sibling whose parent split has a Rewind Point. Non-hideable, non-drop-target; clicking Rewind re-flies the sibling from the moment of the split.
- **Append-only supersede** — merging a re-fly appends a `RecordingSupersede(old, new)` relation. The original recording is never mutated or deleted; ghost/claim subsystems filter via the relation list. Tree invariant preserved (additive only).
- **Narrow v1 supersede scope** — the only ledger retirement on supersede is `KerbalDeath` plus reputation penalties bundled with it; kerbals return to active via the normal reservation walk. Contracts, milestones, facilities, strategies, tech, science, and other rep/funds deltas stay sticky in career totals.
- **Crashed re-fly stays rewindable** — merging a crashed attempt commits as `CommittedProvisional` and remains an Unfinished Flight the player can try again. Merging a stable outcome seals the slot as `Immutable`.
- **Effective-state model** — `EffectiveRecordingSet` and `EffectiveLedgerSet` give every subsystem one rule for "what counts right now." A narrow session-suppressed subtree carve-out applies during an active re-fly only to physical-visibility subsystems (ghost walker, claim tracker, map / tracking-station / CommNet). Career state reads ERS/ELS directly.
- **Journaled staged merge** — irreversible file operations (deleting reap-eligible Rewind Point quicksaves) happen only after a durable save, and the nine-phase `MergeJournal` recovers the merge on the next load if a crash occurs mid-sequence.
- **Revert-during-re-fly dialog** — intercepts stock Revert-to-Launch while a session is active and offers Retry from Rewind Point / Discard Re-fly / Continue Flying.

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v0.9.1 — Unfinished Flights Stable Leaves & Re-Fly Hardening

The 0.9.1 line shipped the Phase 12.5 Unfinished Flights stable-leaf extension and a focused Re-Fly hardening pass. Full design (now folded into the unified Rewind-to-Separation spec): `docs/parsek-rewind-to-separation-design.md` . Pre-implementation spec for the stable-leaf extension is archived at `docs/dev/done/parsek-unfinished-flights-stable-leaves-design.md` ; research note (R17) at `docs/dev/research/extending-rewind-to-stable-leaves.md` .

- **Broader Unfinished Flights predicate** — `IsUnfinishedFlight` now includes controllable non-focus Rewind Point children that end `Orbiting` or `SubOrbital` , plus stranded EVA kerbals with non-boarded terminal states. Focus-continuation upper stages, debris, and successful auto-recovered boosters stay out of the list.
- **STASH system group and row actions** — the virtual Unfinished Flights group is now displayed as `STASH` ; rows expose `Fly` and `Seal` , and stable terminal leaves backed by a Rewind Point can be manually `Stash` ed without changing the underlying recording or merge state.
- **Persistent slot signals** — `ChildSlot.Sealed` , `ChildSlot.Stashed` , `RewindPoint.FocusSlotIndex` , and `ReFlySessionMarker.SupersedeTargetId` preserve stable-leaf qualification, manual closure, and linear re-fly chain extension across save/load.
- **Helper extraction** — Unfinished Flight membership and route resolution now live outside the table UI so commit/merge/reap code can use the same predicate as the Recordings Manager.
- **Re-Fly supersede linearization** — repeated re-flies append from the slot's prior effective tip rather than creating origin-rooted star relations. Legacy star-shaped portions are tolerated while new appends form the dominant linear chain.
- **Re-Fly visibility and resume hardening** — session suppression no longer hides unrelated side-off siblings, raw playback/materialization paths share relation-supersede filtering, doubled GhostMap ProtoVessels are suppressed during pending-tree restore windows, and quickload-resume keeps active-only trim scope intact.
- **Ghost trajectory follow-through** — 0.9.1 also folded in the current ghost trajectory rendering hardening: continuous terrain correction, structural-event snapshots, outlier rejection, anchor/co-bubble fixes, and recorded-anchor fallbacks for Re-Fly relative playback.

Phase 13: Logistics (Supply Routes)

Stock-first automated cargo delivery from proven player-flown Supply Runs. Full design: `docs/parsek-logistics-supply-routes-design.md` . Fly a cargo run once, dock, transfer stock cargo, undock, commit, then confirm the prompt to create a recurring Supply Route.

- **v1 route shape** — docking-only, delivery-only, single-stop Supply Routes created from complete dock/transfer/undock Supply Runs. Claw/grapple/crossfeed, pickup routes, crew delivery, multi-stop routes, and round-trip linking are deferred.
- **Stock proof-of-work** — route creation derives the delivery manifest from connection-scoped snapshots: cargo must leave the transport-side part set and appear on the endpoint-side part set during the docking window. Aggregate merged-vessel totals are not enough proof.
- **Origin cost model** — KSC-origin Career routes charge a stock-realistic funds cost for the source vessel parts plus used/delivered cargo. Non-KSC v1 origins require the Supply Run to start docked to a real depot vessel so recurring cargo can be debited from a stock vessel.
- **UT-driven scheduler** — route progress, delivery, pause behavior, save/load catch-up, and time-warp behavior are driven by universal time in `ParsekScenario` , with ghost playback treated as visual evidence rather than authoritative timing.
- **Endpoint resolution** — orbital endpoints use recorded vessel PID only. Surface endpoints prefer the recorded PID and may fall back to one nearest compatible stock vessel near the recorded coordinates, never to an abstract area warehouse.

- **Visual presence** — ghost supply vessels replay the recorded chain when visuals are available, but route execution is pure math: deduct at origin, wait, add to destination. No physical vessel is spawned during transit.

Logistics prerequisites added to Phase 13: Phase 11 provides base vessel-level resource and inventory manifests. Supply Routes also require connection-scoped capture extensions: dock/undock resource and inventory manifests by transport/endpoint part PID set, `TransferTargetVesselPid`, `TransferKind`, `TransferEndpointSituation`, `StartDockedOriginVesselPid`, and exact `InventoryPayloadItem` snapshots from canonical `STOREDPART` data so inventory deliveries preserve per-item state.

Every supply ship remains a replay of a real mission the player flew, but v1 deliberately keeps the mechanics narrow so the first implementation is reliable and stock-realistic.

Phase 14: Cooperative Async Multiplayer

Multiple players contribute recordings to a shared timeline. The Kerbal system feels populated with vessels flying and bases being built. All players share one game actions timeline — science, funds, reputation, contracts, and kerbals are pooled.

Gloops Extraction (Phase 14 Prerequisite)

Extract the ghost playback engine into a separate assembly (`Gloops.dll`) within the same repository. This provides build-time boundary enforcement and defines the `.gloop` file format needed for recording export/import. See `docs/dev/gloops-recorder-design.md` for the full design.

Extraction timing rationale: The engine boundary already exists (IPlaybackTrajectory, IGhostPositioner, GhostPlaybackEngine with zero Recording references). Extracting earlier than Phase 14 adds overhead without user benefit — new features through Phase 13 still touch engine code, and doing that across assemblies adds friction. Phase 14 is the natural trigger because recording export/import requires a standalone file format (`.gloop`), which is exactly what the extraction produces.

Extraction scope:

- Separate .csproj in the same repo (not a submodule yet)
- 17 files move to Gloops, 2 files need pre-extraction splitting
- Pre-extraction refactors: split `GhostPlaybackLogic.cs` (engine vs. policy), extract recorder from `FlightRecorder.cs`, `ParsekLog` abstraction
- Parsek becomes a consumer of the Gloops API

Standalone Gloops mod (post Phase 14, if demand exists):

- Split into separate repository / submodule
- Content pack system for ambient world activity (KSC traffic, scenery)
- Standalone UI (loop manager, pack toggles, settings)
- Custom mesh support for non-vessel content

Recording Export/Import

Prerequisite for multiplayer. Share recordings as standalone `.gloop` archive files. Export bundles trajectory data, vessel snapshot, and metadata. Import validates, assigns new IDs, and injects into current save. Handles missing parts gracefully (warn, skip, or substitute).

Shared Timeline

- **Shared recordings folder** — local or cloud-synced where players drop recording files

- **Import/merge on load** — Parsek scans the shared folder for new recordings and merges them into the local timeline as ghosts; purely additive, no conflict with live game state
- **Player identity** — each recording tagged with a player ID for filtering, color-coding, and attribution
- **Launch pad allocation** — different players claim different launch sites (Phase 10); each player's recordings originate from their claimed sites
- **Version tolerance** — additive recording format means players on different Parsek versions can share recordings safely

Players never need to be online simultaneously. Fly missions, export recordings. Others import whenever they play. The timeline converges over time — correspondence chess, not real-time multiplayer.

Phase 15: Competitive Play & Space Race

Player boundaries that enable competitive multiplayer. Each player has their own game actions timeline (separate science, funds, reputation, contracts, kerbals). Only things with visible effect on the common game world are shared — vessel recordings play as ghosts in everyone's game, but each player's economy is independent.

Player Boundaries

- **Per-player game actions timeline** — each player's resource ledger is independent; importing another player's recordings adds their ghosts to the world but does not affect the local player's science, funds, or reputation
- **Shared world layer** — vessel recordings (trajectories, part events, ghost chains) are shared and visible to all players; the physical world is common, the economic world is separate
- **Milestone visibility** — all players can see each other's milestones (first orbit, first Mun landing) but milestones only feed into the achieving player's resource pipeline

Space Race

- **Pre-built recording packs** — a set of recordings representing an AI space program's progression (first orbit on Day 30, Mun landing on Day 150, etc.); hand-crafted or community-contributed
- **Milestone racing** — Parsek already tracks milestones; space race is a comparison UI showing both programs' achievements side by side
- **Difficulty scaling** — different packs for different skill levels
- **Community races** — players export their full campaign as a space race opponent ("Can you beat Scott Manley's career run?")

The Dependency Chain

```
v0.3–v0.5: Core Recording System (COMPLETE)
    │  Recording, playback, loops, ghost chains, spawn safety,
    │  time jump, rendering zones, multi-vessel sessions
    │
    ▼
v0.6: Game Actions System (COMPLETE)
    │  Science, funds, reputation, kerbals, contracts,
    │  facilities, strategies, reservations, recalculation
    │
    ▼
Phase 9: Timeline (v0.7 ✓)
    │  Unified chronological view of all committed events,
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    │  significance tiers, filtering, rewind from timeline
    │
    ▼
Phase 10: Location Context (v0.7 ✓)
    │  Recordings know WHERE they start/end (body, biome, situation)
    │
    ▼
Phase 11: Resource Snapshots (done)
    │  Recordings know WHAT they carry
    │
    ▼
Phase 11.5: Recording Optimization & Observability (v0.8.x)
    │  Observability + ghost LOD + trajectory/snapshot shrink shipped;
    │  remaining follow-up is synthetic stress benchmarking/tuning
    │
    ▼
Phase 12: Rewind to Separation (v0.9 ✓)
    │  Rewind Points at multi-controllable splits, Unfinished Flights
    │  group, append-only supersede, narrow v1 scope. Independent of
    │  Phase 13 — both consume Phase 11 resource/inventory/crew manifests.
    │
    ▼
Phase 12.5: Stable Leaves + Re-Fly Hardening (v0.9.1 ✓)
    │  STASH, Fly/Seal/Stash, slot-aware stable leaves, linear supersede,
    │  quickload/visibility/relative-playback hardening
    │
    ▼
Phase 13: Logistics (Supply Routes)
    │  Stock-first Supply Routes from proven dock/transfer/undock Supply Runs
    │
    ▼
Gloops Extraction ─── Extract ghost engine to separate assembly,
    │                   define .gloop file format
    ▼
Phase 14: Cooperative Async Multiplayer
    │  .gloop export/import, shared folder, player identity
    │
    ▼
Phase 15: Competitive Play + Space Race
    │  Per-player game actions, milestone racing, opponent packs
    │
    ▼
Phase 16: Mod Compatibility
    │  Kerbal Konstructs, KSC Switcher, Extraplanetary Launchpads,
    │  off-world construction, modded launch sites via reflection
    │
   ...
Standalone Gloops Mod (if demand exists)
       Separate repo, content packs, standalone UI, custom meshes
```

Phase 16: Mod Compatibility

All prior phases target stock KSP (including Making History DLC). This phase adds support for popular mods via reflection-based detection and API integration. Only attempted after the stock experience is stable and fun.

- **Kerbal Konstructs / KSC Switcher** — detect custom launch sites, capture site name at recording start, display in timeline and Recordings Manager
- **Extraplanetary Launchpads** — detect EL-spawned vessels (builder PartModules: `ELLaunchpad`, `ELDisposablePad`, `ELSurveyStation`), capture pad name, handle empty `launchedFrom` field
- **Off-world construction mods** — detect vessels built outside KSC, tag recordings with construction origin
- **Mod detection pattern** — `AssemblyLoader.loadedAssemblies` (already used for PersistentRotation, RemoteTech). Reflection for API access. Graceful degradation when mods are absent.
- **Compatibility testing** — CustomBarnKit, Strategia, Contract Configurator (see T43)

Deferred

Planetarium.right drift compensation for long orbital segments

KSP's inertial reference frame may drift over very long time warp durations. This could cause ghost orientation mismatch for interplanetary transfer segments. Needs empirical measurement first — if drift is sub-degree for typical segment lengths, no fix needed.

Crash-safe pending recording recovery (partial)

Commit crash window closed (sidecar files flushed immediately). Remaining gap: stashed-but-not-yet-committed recordings (merge dialog pending) still live only in RAM. Solution: write a pending manifest file when a recording is stashed, auto-recover on next load if the recording wasn't committed or discarded. Low priority — remaining gap is stash→commit only.

Additional part event coverage

- Two-phase engine startup (spool-up animations on some engines)
- Engine plate covers / interstage fairings (partially fixed — mesh cloned but positioning may be wrong)

Ghost LOD follow-up

Distance-based ghost LOD shipped in `0.8.1`, including the hidden-tier ghost unload/rebuild follow-up. Particle pooling for engine/RCS FX is not scheduled; the Phase 11.5 outcome there was observability/measurement only. Storage-side optimization is also now shipped, so the remaining Phase 11.5 follow-up is synthetic stress benchmarking/tuning.

Kerbal reservation refactor (T44)

Replace `rosterStatus = Assigned` workaround with Parsek-internal state + Harmony crew dialog filtering. Would eliminate 2 workaround patches and ~27 KSP warnings per session. Low priority — current workaround is functional.

Hyperbolic escape orbit line rendering (189b)

Ghost escape orbits clip at finite distance (~12,000 km). Active vessels show full escape trajectory via PatchedConicSolver which ghosts don't have. Needs custom LineRenderer or orbit line extension.

Out of Scope

- **Racing modes or lap timing**
- **AI playback or autopilot**
- **Real-time multiplayer synchronization** — Parsek's multiplayer model is async (Phases 13–14). No shared physics simulation, no lockstep networking.
- **Taking control of recorded vessels mid-playback** — jumping into a live ghost while it is playing out creates unresolvable paradoxes (recording future events, reserved crew, applied resource deltas). The complexity is not worth the payoff. **Note:** the narrower "Rewind to Separation" feature (see above) does allow re-flying a sibling vessel from a past split event — that works because it rewinds UT to the split moment and replaces the sibling's BG-crash via append-only supersede, rather than hijacking an in-flight ghost.
- **Timeline branching or alternate histories**